

SIMATIC HMI

WinCC Customer controls

System Manual

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Printing the Online Help

Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

DANGER

indicates that death or severe personal injury **will** result if proper precautions are not taken.

WARNING

indicates that death or severe personal injury **may** result if proper precautions are not taken.

CAUTION

indicates that minor personal injury can result if proper precautions are not taken.

NOTICE

indicates that property damage can result if proper precautions are not taken.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

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Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be complied with. The information in the relevant documentation must be observed.

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Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

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Overview (RT Advanced, RT Professional)

Use of WinCC Control Development

WinCC Control Development can be used without a license.

You can find the current version of WinCC Control Development for download in the Siemens Industry Online Support, entry 109760182 "Control Development Kit" (<https://support.industry.siemens.com/cs/de/de/view/109760182>).

Introduction

User-specific controls are used to create and standardize complex, frequently used screen objects. The "WinCC control development" add-on is required to develop controls. WinCC Control Development can be used without a license. You require Visual Studio 2010 or higher and .NET Framework 4.0 as the programming environment.

You can, for example, place several motors of a similar type in a plant. In a customer control, you can summarize motor data such as operating mode, setpoint, actual value, rotational speed, operational status, frequency.

M

Value

Process value

Setpoint

Frequency

Hz

Mode

Manual

Auto

Operating mode

On

Off

Speed

1/minute

When the customer control is finished, you can integrate it an unlimited number of times into WinCC screens, as well as in other applications.

Central changes to the customer control are automatically adopted at all usage locations. In this way, you can later expand a customer control with additional functionality. The changes are effective on all customer controls of this type used in WinCC screens that are already configured.

If a customer control has its own property page, these properties can also be configured directly in WinCC. You open the property page of the control using the shortcut menu in WinCC.

Required WinCC edition

You can only use customer controls with the WinCC editions "WinCC Advanced" and "WinCC Professional".

Target group of this documentation

This documentation is aimed at configuration engineers with knowledge of VB.net or C#.

Service & Support on the Internet

In addition to our documentation, we offer a comprehensive knowledge base on the Internet at: www.siemens.com/automation/service&support

System requirements

Please note the system requirements for your WinCC edition as well as Visual Basic 2010 or higher with .Net 4.0. Otherwise, the system requirements for Visual Studio and WinCC apply.

Demo project (RT Advanced, RT Professional)

Introduction

The "WinCC Control Development" add-on contains demo projects. The following documentation shows how you can develop and test the customer control "MyControl" using these examples.

Visual Studio demo projects

The demo projects include the interface to the data provider of WinCC. The various ITag-Interfaces methods are used.

The demo projects are programmed in Visual Studio 2010 and use the .net Framework 4.0 Client Profile.

The demo project for Visual Studio includes the class library "CCITagTestLibrary". It contains the following customer control:

The screenshot displays a WinCC control interface with eight distinct method panels arranged in a grid. Each panel contains input fields and a 'Start' button:

- ReadTagCyclic:** Includes fields for TagName, UpdateRate, and Cookie, along with a radio button and 'Start' and 'Cancel' buttons.
- ReadTag:** Includes fields for TagName and Value, and a 'Start' button.
- ReadTagAsync:** Includes fields for TagName, Cookie, and Cache, and a 'Start' button.
- WriteTag:** Includes fields for TagName and Value, and a 'Start' button.
- WriteTagAsync:** Includes fields for TagName, Cookie, and Value, and a 'Start' button.
- OnDataChanged:** Includes fields for TagName, Cookie, State, and Value.
- OnError:** Includes fields for TagName, Cookie, and Error.
- OnWriteComplete:** Includes fields for TagName and Cookie.

The demo project for Visual Studio includes the test form "CCITagApplication. You use it to test your own customer controls.

Note

Property pages in the customer control

We do not recommend creating any controls with property pages in Visual Studio.

Principle

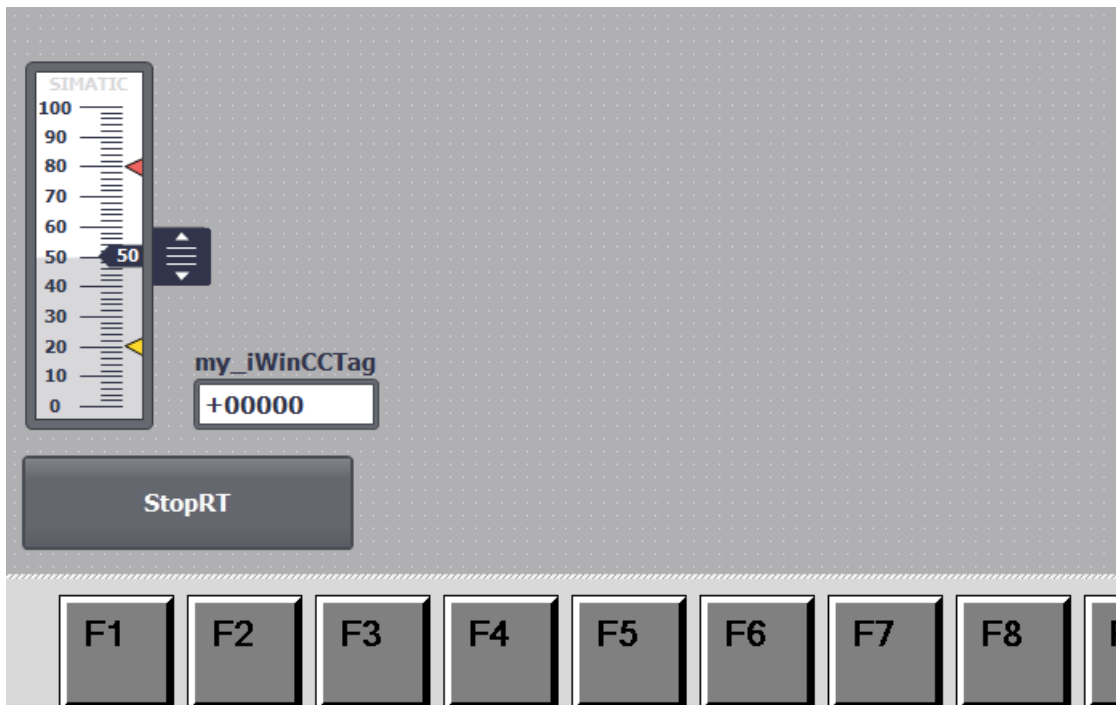
Task

In this documentation, you create the following extract from the demo project as your own customer control ("MyControl").

The control is used to get the current value of a tag. Enter the required tag name in runtime under "TagName". When you click on "Start", the control displays the current value of the tag under "Value".

The screenshot displays a collection of WinCC controls arranged in a grid. The 'ReadTag' control, located in the top-middle position, is highlighted with a red rectangular border. It contains two text input fields: 'TagName' and 'Value', and a 'Start' button below them. To its left is the 'ReadTagCyclic' control, which includes 'TagName', 'UpdateRate', 'Cookie' fields, a radio button, and 'Start' and 'Cancel' buttons. To its right is the 'ReadTagAsync' control, featuring 'TagName', 'Cookie', 'Cache' fields and a 'Start' button. Below 'ReadTag' is the 'WriteTag' control, with 'TagName', 'Value' fields and a 'Start' button. To its right is the 'WriteTagAsync' control, with 'TagName', 'Cookie', 'Value' fields and a 'Start' button. At the bottom of the grid are three more controls: 'OnDataChanged' (with 'TagName', 'Cookie', 'State', 'Value' fields), 'OnError' (with 'TagName', 'Cookie', 'Error' fields), and 'OnWriteComplete' (with 'TagName', 'Cookie' fields).

You will then configure all required objects in WinCC in order to test your control in Runtime.



See also

Creating and testing customer controls (Page 13)

Creating and testing customer controls (RT Advanced, RT Professional)

3

Introduction

This section shows the basic procedure for creating and testing a customer control in Visual Studio.

Use the supplied class library "CCITagTestLibrary" as a template to follow the steps for configuration.

Requirements

- Visual Studio 2010 or higher is installed.
- WinCC RT Advanced or WinCC RT Professional is installed.

Basic procedure

For the customer control "MyControl" class library:

1. Create Visual Studio project (Page 15)
 - Create class library
 - Create user structure element
 - "Reference Siemens.Runtime.ControlDev.dll"
2. Create a user interface (Page 19)
3. Create program code (Page 21)
4. Create customer control (Page 23)
5. Test customer control in Visual Studio (Page 28)
6. Test customer control in WinCC (Page 30)

See also

Demo project (Page 9)

Creating customer controls (Page 15)

Testing customer controls (Page 25)

Creating customer controls (RT Advanced, RT Professional)

4.1 Creating Visual Studio project (RT Advanced, RT Professional)

Introduction

You create your own customer control in the demo project folder supplied.

To access the data provider of WinCC, create a reference in your project to the "Siemens.Runtime.ControlDev.dll" library.

Requirement

Visual Studio 2010 or higher is installed.

Opening demo project

To create a new Visual Studio project for your customer control in the project folder, proceed as follows:

1. Navigate to <Program Files Folder>\Siemens\Automation\Control_Development_*\samples\VB100
The "CCITagTest" folder contains the Visual Studio project "CCITagTestLibrary".
2. Copy the entire "CCITagTestLibrary" folder to a local destination directory.
3. Double-click on the "CCITagTestLibrary.sln" file.

Visual Studio is opened.

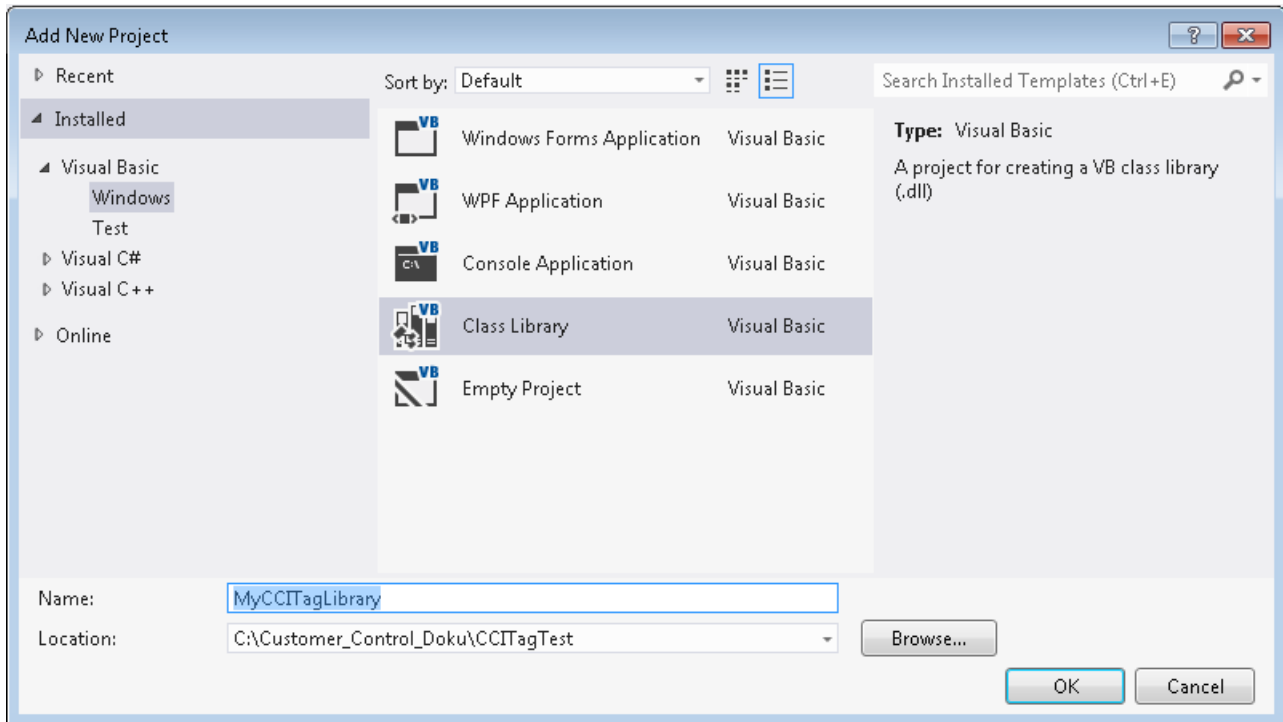
The "CCITagApplication" and "CCITagTestLibrary" projects are displayed in the project folder Explorer.

Creating new Visual Studio project

1. Select "File > Add > New Project".
A dialog opens.
2. Select the default "Class library" under "Templates > Visual Basic > Windows".

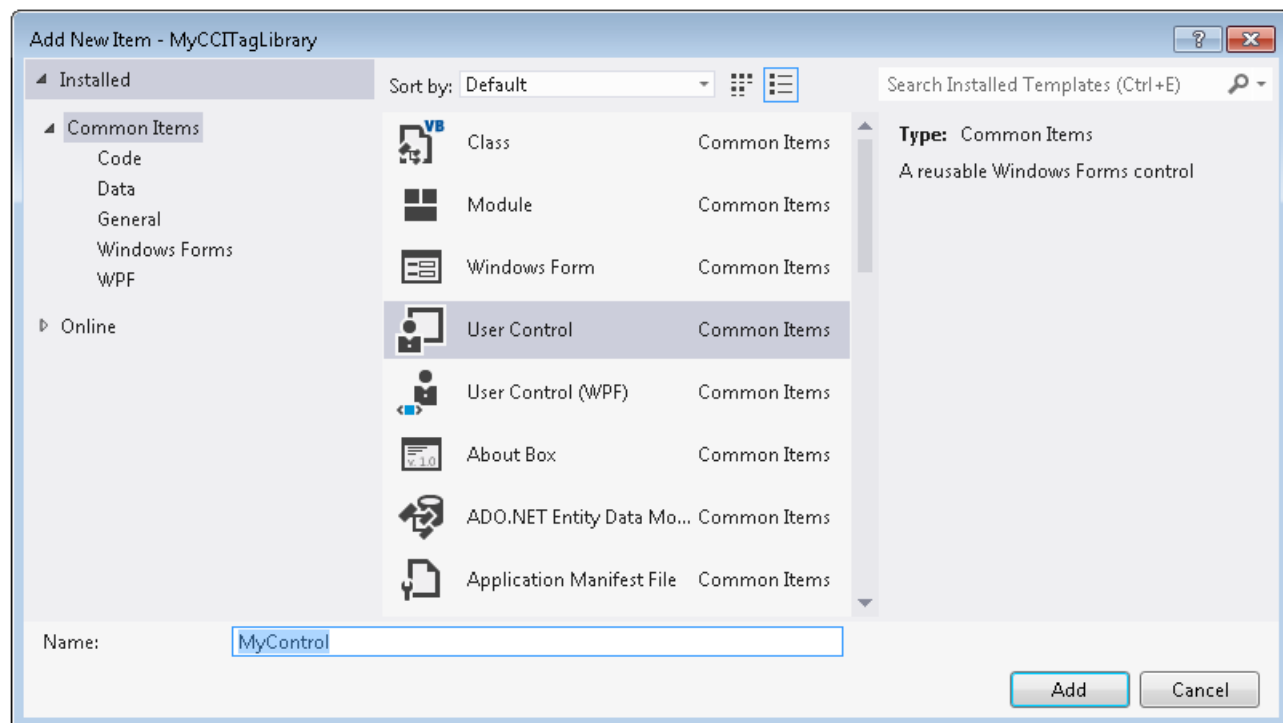
4.1 Creating Visual Studio project (RT Advanced, RT Professional)

3. Enter "MyCCITagLibrary" for the name and click "OK".



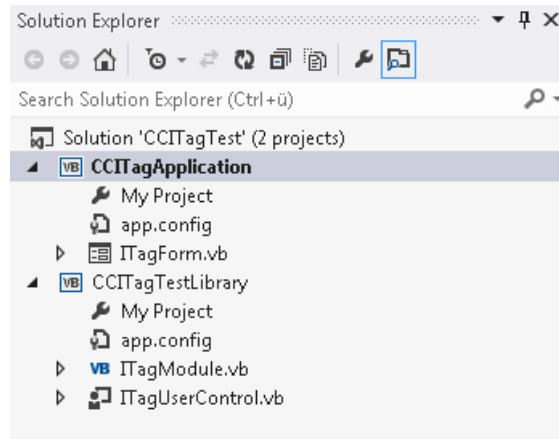
4. Select "Add > User control element" in the shortcut menu of the new class library.

5. Enter "MyControl" for the name and click "Add".



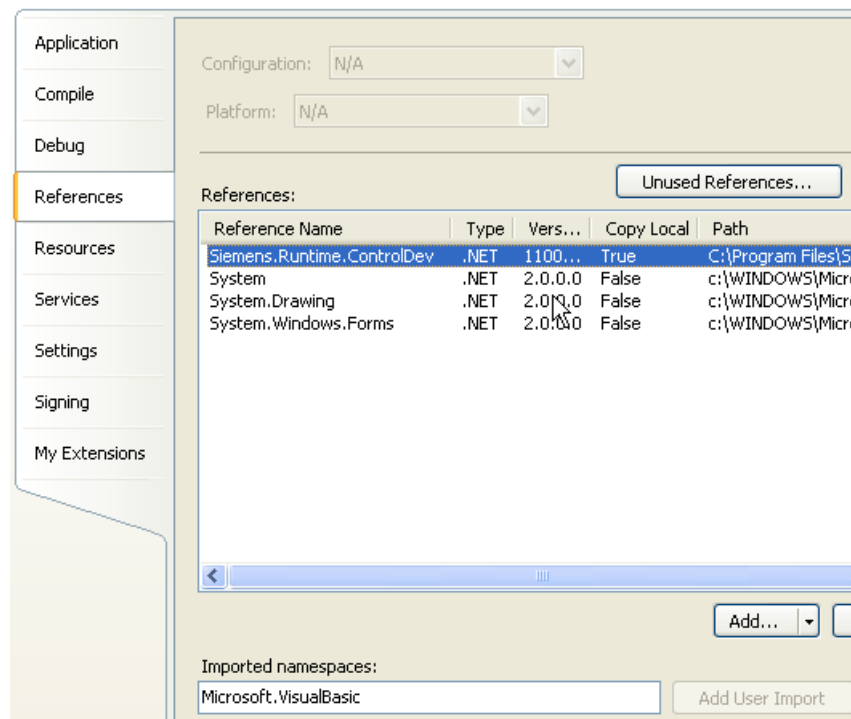
Set reference to "Siemens.Runtime.ControlDev.dll"

1. Double-click the "MyCCITagLibrary > My Project" entry in the project folder explorer.



The Project Designer opens.

2. Click on "References" in the Project Designer.



3. Under "..\CCITagTestLibrary\bin", load the "Siemens.Runtime.ControlDev.dll":
4. Check whether the reference is also set in the projects "CCITagApplication" and "CCITagTestLibrary".

Note

.net version

Make sure that ".Net Framework 4.0 Client Profile" is also set in the newly created Visual Studio project.

See also

Creating a user interface (Page 19)

4.2 Creating a user interface (RT Advanced, RT Professional)

Settings

For the configuration, you require objects with the following settings:

Objects		Property	Setting	Use
	GroupBox	Text	ReadTag	-
	Label	Text	TagName	-
		Text	Value	-
	Textbox	(Name)	txtReadTagTagName	Entering a tag name
		(Name)	txtReadTagValue	Displaying a tag value
		ReadOnly	True	
	Button	Text	Read	Reading a tag value
		(Name)	btnReadTagStart	

The objects required are also included in the "GroupBox5" of the demo project.

Requirement

- The demo project is opened with the newly created user structure element "MyControl".
- The toolbox is displayed.
- The properties window is displayed.

Procedure

To create the user interface, follow these steps:

1. Double-click the "MyCCITagTestLibrary > MyControl.vb" entry in the project folder explorer. The Designer opens.
2. Insert the relevant objects into the form from the toolbox and configure their properties and events.

Note

Usable events

Events that include the character sequence "Changed" are not displayed in WinCC.

Configure only events that do not use the addendum "Changed".

Result

The user interface for the "MyControl" customer control has been created.



See also

Creating program code (Page 21)

4.3 Creating program code (RT Advanced, RT Professional)

Introduction

For access to the WinCC tag values, WinCC provides interfaces for access to the WinCC data provider using various methods. The connection to the WinCC data provider is established or terminated using methods.

To use customer control to access the WinCC tag values, implement the ITag interface in your Visual Studio project. You use the following ITag methods

- **Register**
The "Register" method is used to register your control with the data provider. Access is identified by a cookie.
- **ReadTag**
The "ReadTag" method is used for reading the tag value of the WinCC tag "my_IWinCCTag" at the same time.

Requirement

- A Visual Studio project "MyCCITagLibrary" has been created
- "Siemens.Runtime.ControlDev.dll" class library is referenced.

Procedure

To create the program code, proceed as follows:

1. Double-click "MyITagUserControl.vb" in the project folder Explorer.
2. Define the required tags and the ITag interface in the class properties. To do this, generate an ITag object from the "Siemens.Runtime.ControlDev.dll".

```
Dim m_Tagname As String ' tag name
Dim m_Delimiter As String ' Delimiter between the tag name and the tag member
Dim m_Active As Boolean ' Activate/deActivate the Tags connection
Dim m_RegisterCookie As Long

Dim ActualViewFlag As Integer ' local variable to hold the actual view flag
Dim ActualViewFrame As Integer ' local variable to hold the actual view frame displayed

' ITag interface
Dim m_ITag As ITag
```

3. Insert the following Event Handler . You can also use the sub-procedure "ITagUserControl_Load" from the demo project.

```
Private Sub ITagUserControl_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MyBase.Load

    m_Delimiter = "."
    m_Active = True
    ActualViewFlag = 1

    ' Create the Tag Service
    ChangeTagServer(False)
End Sub
```

4.3 Creating program code (RT Advanced, RT Professional)

4. Reference the interface. You can also use the sub-procedure "ChangeTagServer" from the demo project.
To ensure that your customer control can access the ITag interface, the ProgID of the data provider must be transferred.
 - ProgID for WinCC Runtime Professional: "SimaticHMI.CoRtTagDataProvider.1"
 - ProgID for WinCC Runtime Advanced: "FwTagDataProvider.FwTagDataProvider"

```
Private Sub ChangeTagServer(ByVal bRemove As Boolean)
    ' At runtime change the Data Server only if the Active property is set
    If Not DesignMode And m_Active Then
        On Error Resume Next

        ' if nested in WinCC: get the Tagset Object from the container
        m_ITag = Site.GetService(GetType(ITag))

        ' or if nested in Windows Form
        If m_ITag Is Nothing Then

            m_ITag = CreateObject("SimaticHMI.CoRtTagDataProvider.TagDataProvider.1")

        End If

        ' Connect to server
        m_RegisterCookie = m_ITag.Register(Me)
        ' Add all items in the current view
        'AddItemsToGroup()
    End If
End Sub
```

5. Double-click the "btnReadTagStart" button in the Designer.
The code window for the "btnReadTagStart_click" sub-procedure opens.
6. Add the following code. You can also use the sub-procedure "btnReadTagStart_Click" from the demo project.

```
Private Sub btnReadTagStart_Click(sender As Object, e As EventArgs) Handles btnReadTagStart.Click
    Try
        Dim TagName As Object = txtReadTagTagName.Text
        Dim Values As Object = m_ITag.ReadTag(m_RegisterCookie, TagName)
        Dim Value
        For Each Value In Values
            txtReadTagValue.Text = CStr(Value)
        Next
    Catch ex As Exception
        MsgBox("ReadTag:" & vbCrLf & ex.Message)
    End Try
End Sub
```

Result

Access to the ITag interface is implemented and the code for the "btnReadTagStart" button is created.

See also

Creating customer controls (Page 23)

Interfaces (Page 33)

4.4 Creating customer controls (RT Advanced, RT Professional)

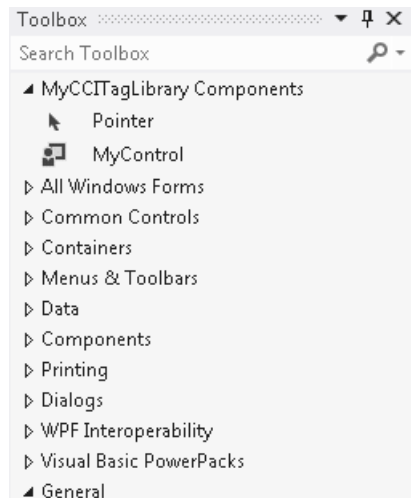
Procedure

To create the customer control, proceed as follows.

1. Double-click the "My Project" entry in the project folder explorer. The Project Designer opens.
2. Click on "Compile" in the Project Designer.
3. Specify the destination directory for the DLL file under "Build output path".
4. Save the change in the Visual Studio project "MyCCITagLibrary".
5. Select the ""MyCCITagLibrary" entry in the project folder explorer. Select the "Create" command in the shortcut menu.

Result

The "MyControl.dll" file is now located in the destination directory. The customer control "MyControl" is displayed in the toolbox.



Testing customer controls (RT Advanced, RT Professional)

5.1 Basic principles for testing customer controls (RT Professional)

Introduction

To test the functionality of the newly created customer control, use the test form "CCITagApplication" in the demo project.

Note

Testing with the test form only works for Runtime Professional.

Basic procedure

To test the customer control, proceed as follows:

1. Open WinCC project for Runtime Professional
2. Add customer control to the Visual Studio test form "CCITagApplication"
3. Test customer control in Visual Studio
4. Add customer control to WinCC
5. Test customer control in Runtime Professional

Note

Property pages in the customer control

We do not recommend creating any controls with property pages in Visual Studio.

See also

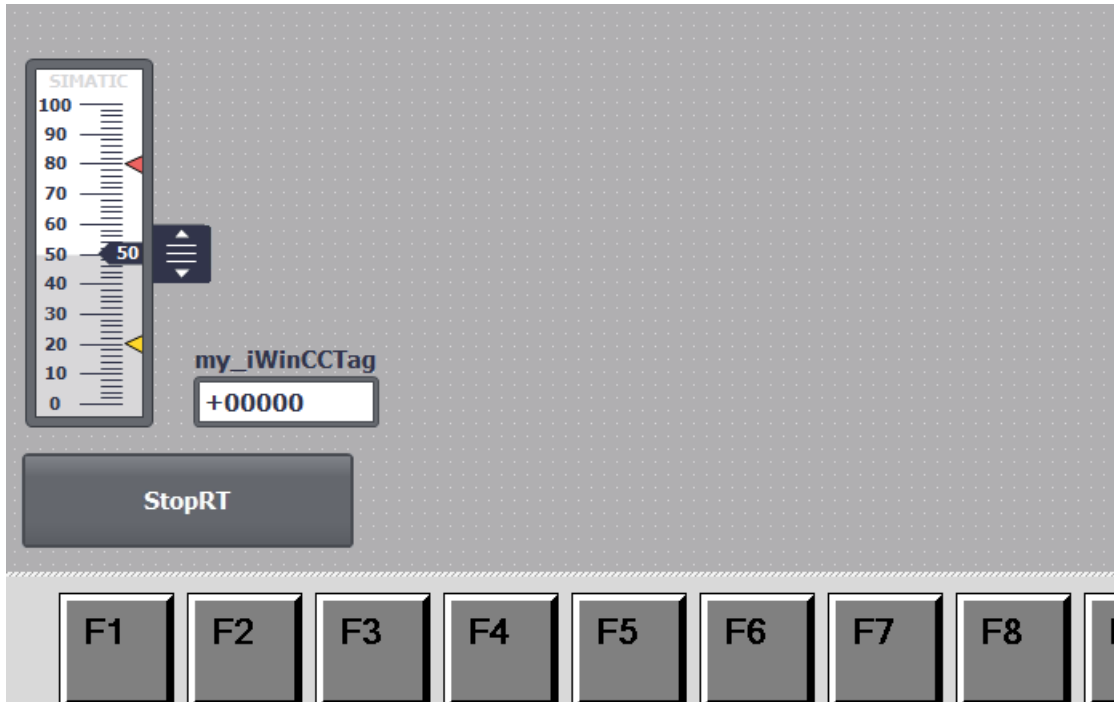
Testing customer controls in Visual Studio (Page 28)

Testing customer controls in WinCC (Page 30)

5.2 Creating a WinCC test project (RT Advanced, RT Professional)

Introduction

To test the customer control "MyControl", you require a WinCC project with a "my_iWinCCTag" tag and a "CC_ITagUserControl" screen.



Procedure

1. Create a new project
2. Create a new WinCC Runtime Professional device.
3. Create an internal "my_iWinCCTag" tag of the "Integer" data type.
4. Create a "CC_ITagUserControl" screen.
5. Create a slider and interconnect it with the "my_iWinCCTag" tag.
6. Configure the "SetTag" system function at the "Change" event of the slider with the following settings:

Parameter	Setting
Tag (output)	my_iWinCCTag
Value	my_iWinCCTag

7. Create an IO field and interconnect it with the "my_iWinCCTag" tag.
8. Create a button.

9. Configure the system function "StopRuntime" at the "Click" event of the button.
10. Start runtime.

Result

The "CC_ITagUserControl" screen is displayed. The tag value in the I/O field will be displayed when you operate the slider.

5.3 Testing customer controls in Visual Studio (RT Advanced, RT Professional)

5.3.1 Adding customer control to test form (RT Advanced, RT Professional)

Introduction

You need a test form to test your customer control in Visual Studio. The test form is a "Windows Forms Application" for Visual Basic. The demo project for Visual Studio contains the test form "CCITagApplication". "Siemens.Runtime.ControlDev.dll" is referenced in the test form.

Requirement

- "MyControl" customer control has been created.
- The WinCC test project is in Runtime.

Procedure

To add your customer control to the test form, proceed as follows:

1. Double-click "My Project" in the project folder explorer under "CCITagApplication". The Project Designer opens.
2. Connect the class library "MyCCITagLibrary" under "References".
3. Check under "References" that the reference has been set to "Siemens.Runtime.ControlDev.dll".
4. Save the project.

Result

Your customer control "MyControl" is connected to the test form.

See also

Testing customer controls in Visual Studio (Page 28)

5.3.2 Testing customer controls in Visual Studio (RT Advanced, RT Professional)

Requirement

- The customer control is connected in the test form "CCITagApplication".
- The WinCC test project is in Runtime.

Procedure

To test the customer control in Visual Studio, proceed as follows:

1. Select the "CCITagApplication" test form in the project folder explorer.
2. Start the "CCITagApplication" form with <F5>.
3. Enter "my_iWinCCTag" in the "TagName" field.
4. Click "Read".

Result

The "Value" field displays the tag value of "my_iWinCCTag". Move the slider to change the tag value in WinCC. The current tag value will be displayed in the I/O field. Click "Read" to update the tag value in the customer control.

5.4 Testing customer controls in WinCC (RT Advanced, RT Professional)

5.4.1 Adding customer control in WinCC (RT Advanced, RT Professional)

Introduction

To use customer controls in WinCC, you need to add them to the "My Controls" palette.

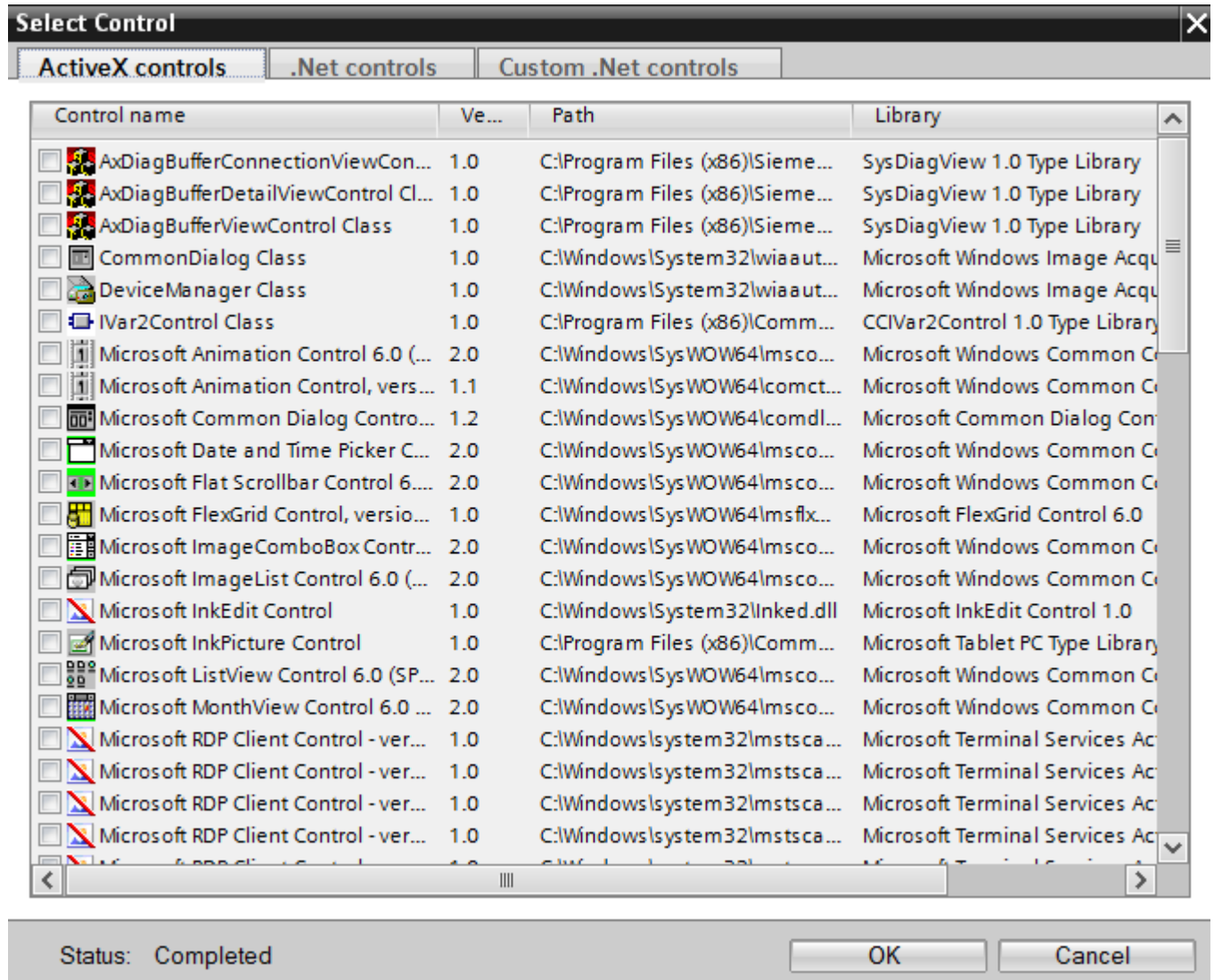
Requirements

- The WinCC test project "CC_ITagUserControl" screen is open.
- The "Tools" task card is open.

Procedure

To add customer controls, proceed as follows:

1. Open the "Toolbox > My controls" task card.
2. Select "Select objects" from the shortcut menu.
A dialog opens. This contains all the controls available on the system. Controls that already exist in the "Toolbox" task card are identified by a check mark.



3. Select the "Specific .Net Controls" tab. Navigate to the required customer control "MyControl" in the "Select assembly" field.
4. Select the "MyControl" customer control.
5. Click "OK".

Result

"MyControl" is displayed in the "Toolbox" task card under "My controls".

See also

Testing customer control in WinCC (Page 32)

5.4.2 Testing customer control in WinCC (RT Advanced, RT Professional)

Requirement

- The "CC_ITagUserControl" screen is opened.
- You have added your control to WinCC. "MyControl" is displayed in the "Toolbox" task card under "My controls".

Procedure

To test the "MyControl" customer control, proceed as follows:

1. Open the "Toolbox > My controls" task card and insert "MyControl" in the screen.
2. Start WinCC Runtime with the start screen "CC_ITagUserControl".
3. Move the slider.
The new value is displayed in the I/O field.
4. Enter "my_iWinCCTag" in the "TagName" field of the control.
5. Click the "Read" button of the control.

Result

The "Value" field displays the tag value that you set on the slider.

Reference (RT Advanced, RT Professional)

6.1 Interfaces (RT Advanced, RT Professional)

Introduction

For access to the WinCC tag values, WinCC provides interfaces for access to the WinCC data provider. The connection to the WinCC data provider is established or disconnected using methods. Methods are also used to read or write tag values synchronously or asynchronously.

- **Synchronous:**
The customer control sends a request to the data provider for which a reply is expected immediately. The customer control waits until a reply comes from the data provider.
- **Asynchronous:**
The customer control sends a request to the data provider for which a reply is not expected immediately. Sending the request and receiving the reply occur at separate times.

Interfaces

- **ITag**
The ITag object supports the following methods:

Method	Description
Register	Registers the customer control as a consumer at the data provider. The consumer receives a cookie from the data provider. The cookie is used to assign a request from the data provider to a consumer.
ReadTag	Reads the value of one or more tags at the same time
ReadTagAsync	Reads the tag value asynchronously
ReadTagCyclic	Cyclic reading of the tag values. The update cycle is used to determine how often the value is updated.
RemoveCyclicTag	Removes one or more tags
Cancel	Cancels the cyclic reading of the tag values
WriteTag	Writes values to one or more tags synchronously
WriteTagAsync	Writes values to one or more tags asynchronously
Unregister	Removes the consumer

- **ITagSink**
The ITagSink object reacts to specific events using methods.

Method	Description
OnCanceled	Occurs when the ITag method "Cancel" has ended.
OnDataChanged	Occurs when a tag value is changed by the following ITag methods: • ReadTagAsync • ReadTagCyclic

Method	Description
OnError	Occurs as an event of an error with the following methods: • ReadTagAsync • ReadTagCyclic • WriteTagAsync • OnDataChanged • OnWriteComplete
OnRemoved	Occurs when the ITag method "RemoveCyclicTag" has ended.
OnWriteComplete	Occurs when the ITag method "WriteTagAsync" has finished.

See also

[ITag methods \(Page 35\)](#)

[ITag methods \(Page 41\)](#)

6.2 ITag methods (RT Advanced, RT Professional)

6.2.1 Register (RT Advanced, RT Professional)

Description

Registers the customer control as a consumer at the data provider. The consumer receives a cookie from the data provider. The cookie is used to assign a request from the data provider to a consumer.

Syntax

RegisterCookie Object.**Register** (CustomerControl)

Parameter

Object

Name of the ITag object

Customer Control

Name of the customer controls

Return value

RegisterCookie

Cookie to identify the consumer

6.2.2 ReadTag (RT Advanced, RT Professional)

Description

Reads the value of one or more tags at the same time.

Syntax

Values Object.**ReadTag**(RegisterCookie, TagNames)

Parameter

Object

Name of the ITag object

RegisterCookie

Cookie to identify the consumer

TagNames

Tag name

Return value

Values

Tag value

6.2.3 ReadTagAsync (RT Advanced, RT Professional)

Description

Reads the value of one or more tags asynchronously.

Syntax

Object.**ReadTagAsync** (RegisterCookie, TagNames, ClientCookies, CacheDevice)

Parameter

Object

Name of the ITag object

RegisterCookie

Cookie to identify the consumer

TagNames

Tag name

ClientCookies

Cookie for identification of tags

CacheDevice

This parameter is only evaluated by WinCC RT Professional.

0 = The tag value is read from the process image (cache).

1 = The tag value is read directly from the AS.

See also

OnDataChanged (Page 41)

OnError (Page 42)

6.2.4 ReadTagCyclic (RT Advanced, RT Professional)

Description

Cyclic reading of the tag values. The update cycle is used to determine how often the value is updated.

Syntax

Object.**ReadTagCyclic** (RegisterCookie, TagNames, UpdateRates, ClientCookies, ServerCookies)

Parameter

Object

Name of the ITag object

RegisterCookie

Cookie to identify the consumer

TagNames

Tag name

UpdateRates

Update cycle of the tags. The value is entered in milliseconds.

ClientCookies

Cookie for identification of tags

Return value

ServerCookies

Cookie for identification of tags The return value is used by the ITag method "RemoveCyclicTag (Page 37)".

See also

OnDataChanged (Page 41)

OnError (Page 42)

6.2.5 RemoveCyclicTag (RT Advanced, RT Professional)

Description

Removes one or more tags.

Syntax

Object.**RemoveCyclicTag** (RegisterCookie, ServerCookies)

Parameter

Object

Name of the ITag object.

ServerCookies

Cookie for identification of tags

See also

ReadTagCyclic (Page 37)

OnRemoved (Page 43)

6.2.6 Cancel (RT Advanced, RT Professional)

Description

Cancels the cyclic reading of the tag values.

Syntax

Object.**Cancel** (RegisterCookie)

Parameter

Object

Name of the ITag object

RegisterCookie

Cookie to identify the consumer

See also

OnCanceled (Page 41)

6.2.7 WriteTag (RT Advanced, RT Professional)

Description

Writes values to one or more tags at the same time.

Syntax

Object.**WriteTag**(RegisterCookie, TagNames, Values)

Parameter

Object

Name of the ITag object

RegisterCookie

Cookie to identify the consumer

TagNames

Tag name

Values

Values written to the tags

6.2.8 WriteTagAsync (RT Advanced, RT Professional)

Description

Writes values to one or more tags asynchronously.

Syntax

Object.**WriteTagAsync**(RegisterCookie, TagNames, ClientCookies, Values)

Parameter

Object

Name of the ITag object

RegisterCookie

Cookie to identify the consumer

TagNames

Tag name

ClientCookies

6.2 ITag methods (RT Advanced, RT Professional)

Cookie for identification of tags

Values

Values written to the tags

See also

[OnError](#) (Page 42)

6.2.9 Unregister (RT Advanced, RT Professional)

Description

Removes the consumer.

Syntax

Object.**Unregister** (RegisterCookie)

Parameter

Object

Name of the ITag object

RegisterCookie

Cookie to identify the consumer

6.3 ITag methods (RT Advanced, RT Professional)

6.3.1 OnCanceled (RT Advanced, RT Professional)

Description

Occurs when the ITag method "Cancel (Page 38)" has ended.

Syntax

Object.**OnCanceled** (RegisterCookie)

Parameter

Object

Name of the ITagSink object

RegisterCookie

Cookie to identify the consumer

6.3.2 OnDataChanged (RT Advanced, RT Professional)

Description

Occurs when a tag is changed by the following ITag methods.

- ReadTagAsync (Page 36)
- ReadTagCyclic (Page 37)

Syntax

Object.**OnDataChanged** (RegisterCookie, TagNames, Values, Qualities, VarStates, TimeStamps, ClientCookies)

Parameter

Object

Name of the ITagSink object

RegisterCookie

Cookie to identify the consumer

TagNames

Tag names

Qualities

Quality code of the tags

VarStates

Tag status of the tags

TimeStamps

Time stamp

ClientCookies

Cookie for identification of tags

See also

OnError (Page 42)

6.3.3 OnError (RT Advanced, RT Professional)

Description

Occurs as an event of an error with the following methods:

- ReadTagAsync (Page 36)
- ReadTagCyclic (Page 37)
- WriteTagAsync (Page 39)
- OnDataChanged (Page 41)
- OnWriteComplete (Page 43)

Syntax

Object.**OnError** (RegisterCookie, TagNames, ClientCookies, Errors)

Object

Name of the ITagSink object

RegisterCookie

Cookie to identify the consumer

TagNames

Tag names

ClientCookies

Cookie for identification of tags

Errors

Error code

6.3.4 OnRemoved (RT Advanced, RT Professional)

Description

Occurs when the ITag method "RemoveCyclicTag (Page 37)" has ended.

Syntax

Object.**Remove**(RegisterCookie, ClientCookies)

Parameter

Object

Name of the ITagSink object

RegisterCookie

Cookie to identify the consumer

ClientCookies

Cookie for identification of tags

6.3.5 OnWriteComplete (RT Advanced, RT Professional)

Description

Occurs when the ITag method "WriteTagAsync (Page 39)" has finished.

Syntax

Object.**OnWriteComplete**(RegisterCookies, TagNames, ClientCookies)

Parameter

Object

Name of the ITagSink object

RegisterCookie

Cookie to identify the consumer

TagNames

Tag names

ClientCookies

Cookie for identification of tags

See also

[OnError \(Page 42\)](#)

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